AP PRECALCULUS

Unit 1 • Worksheet 1.5D • Answer key

Worksheet 1.5D • Polynomial Functions and Complex Zeros

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

A polynomial with real coefficients has zeros 3 and $2i$. What is its least possible degree?

(A) 2
(B) 3
(C) 4
(D) 6

Solution

Answer: (B). $-2i$ is also a zero.

Question 2

NO CALCULATOR

Which polynomial has real coefficients and zeros 1 and i?

(A) $(x-1)(x-i)$
(B) $(x-1)(x^2+1)$
(C) $(x+1)(x^2+1)$
(D) $(x-1)(x^2-1)$

Solution

Answer: (B). $(x-i)(x+i) = x^2+1$.

Question 3 (Free response)

NO CALCULATOR

Write a polynomial in factored form, with real coefficients and least degree, that satisfies the conditions.

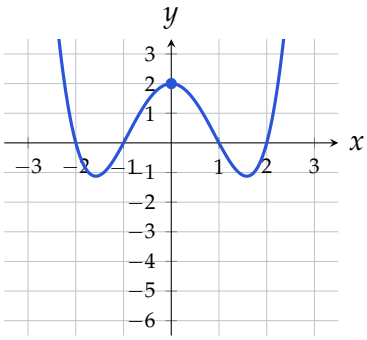
(a) Zeros -2, 0 and 3, leading coefficient 2.
(b) Zeros 1 (multiplicity 2) and $-3i$, leading coefficient 1.
(c) Zeros -1 and 4, the graph touches the axis at $x = 4$, and $p(0) = -32$.

Solution

(a) $2x(x+2)(x-3)$.
(b) $(x-1)^2(x^2+9)$.
(c) $p(x) = a(x+1)(x-4)^2$; $p(0) = a \cdot 16 = -32$, $a = -2$: $p(x) = -2(x+1)(x-4)^2$.

Question 4 (Free response) NO CALCULATOR

The graph of a polynomial p of degree 4 is shown.



Its zeros are $-2, -1, 1, 2$ and $p(0) = 2$. Write $p(x)$.

Solution

$p(x) = a(x+2)(x+1)(x-1)(x-2) = a(x^2-4)(x^2-1)$. $p(0) = 4a = 2$, so $a = \frac{1}{2}$: $p(x) = \frac{1}{2}(x^2-4)(x^2-1)$.